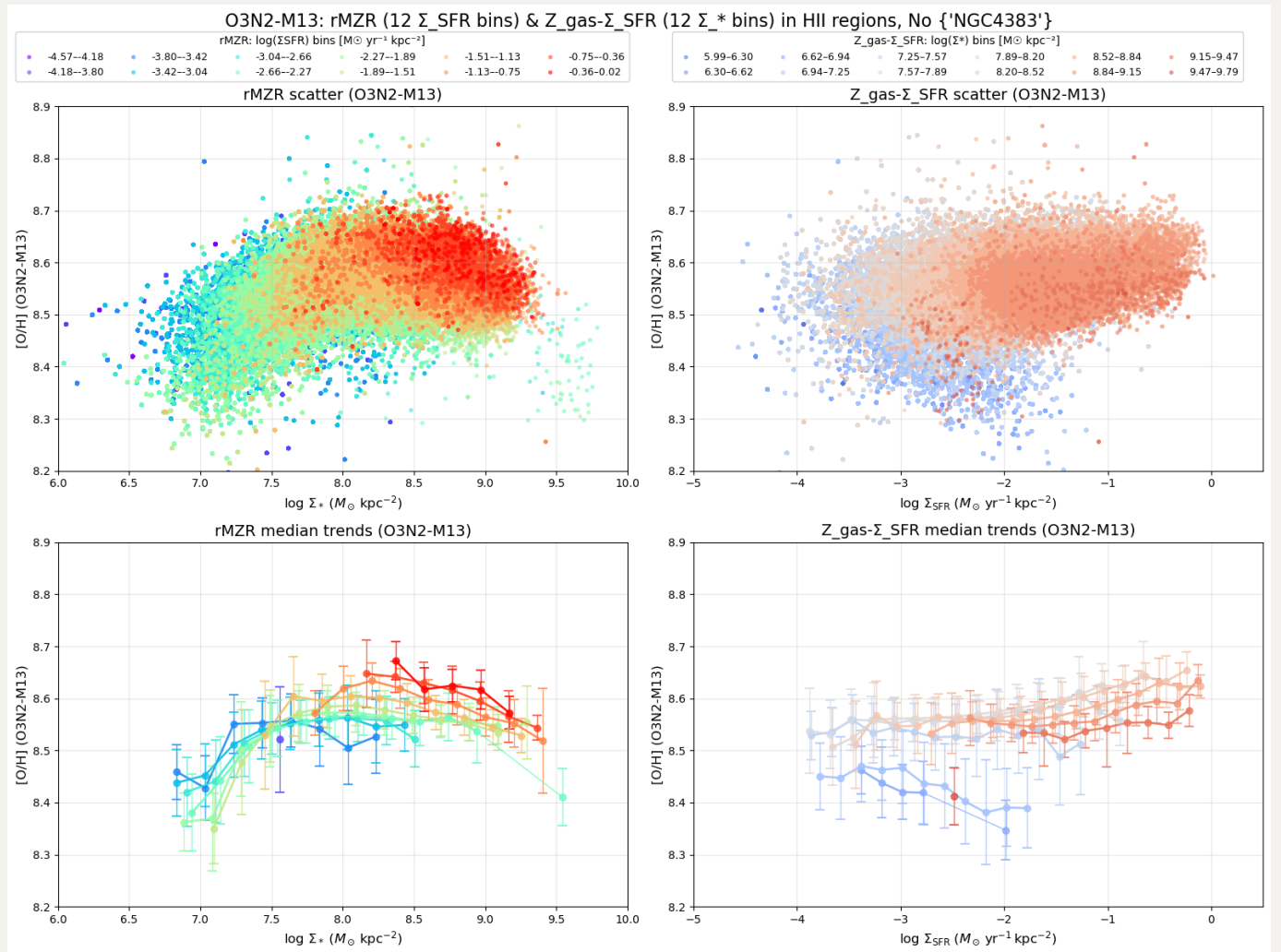


20251013 Offset Map in HII

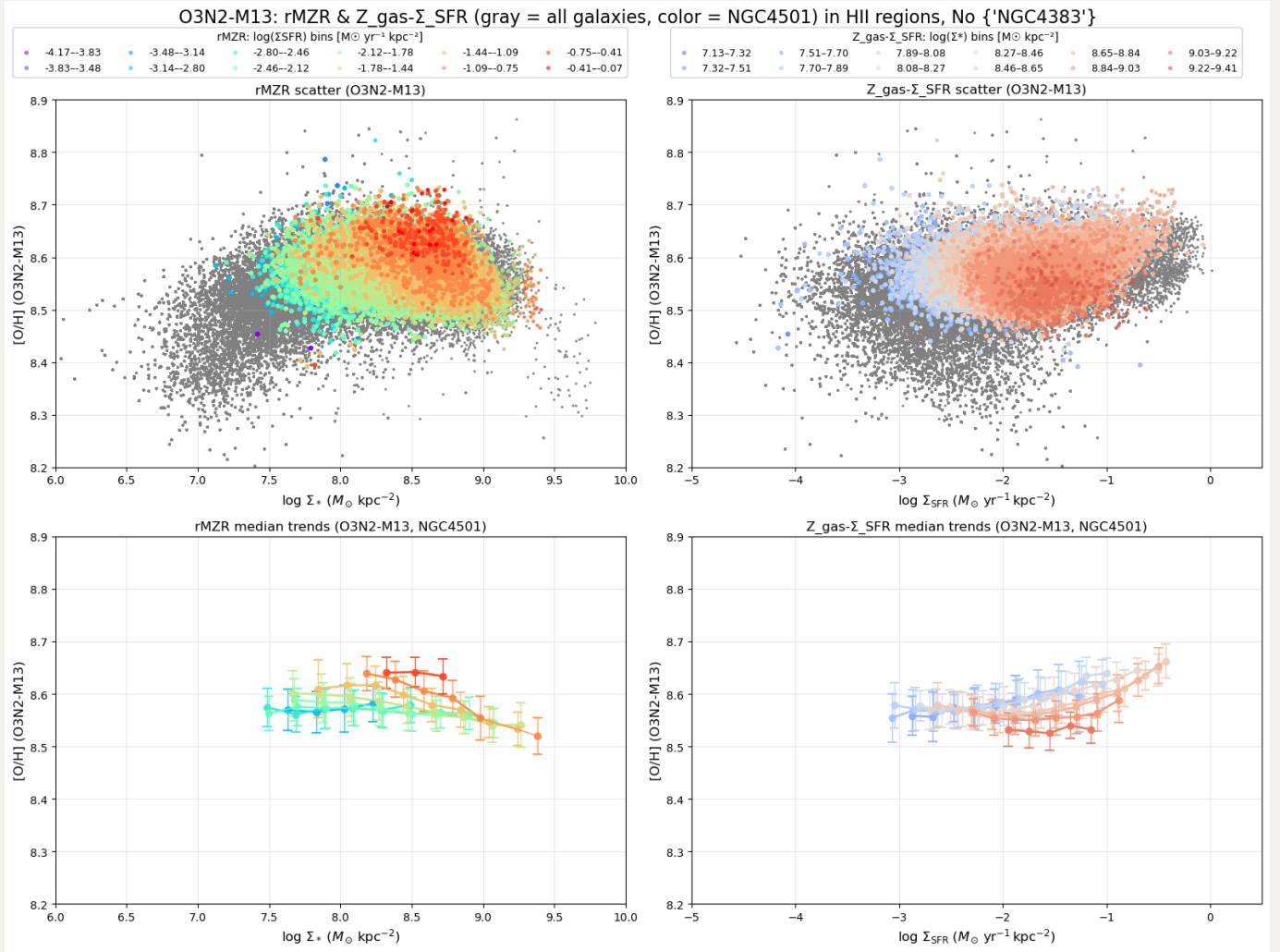
Combined rMZR and $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation in HII regions without NGC4383

Now I update to have 12 bins.

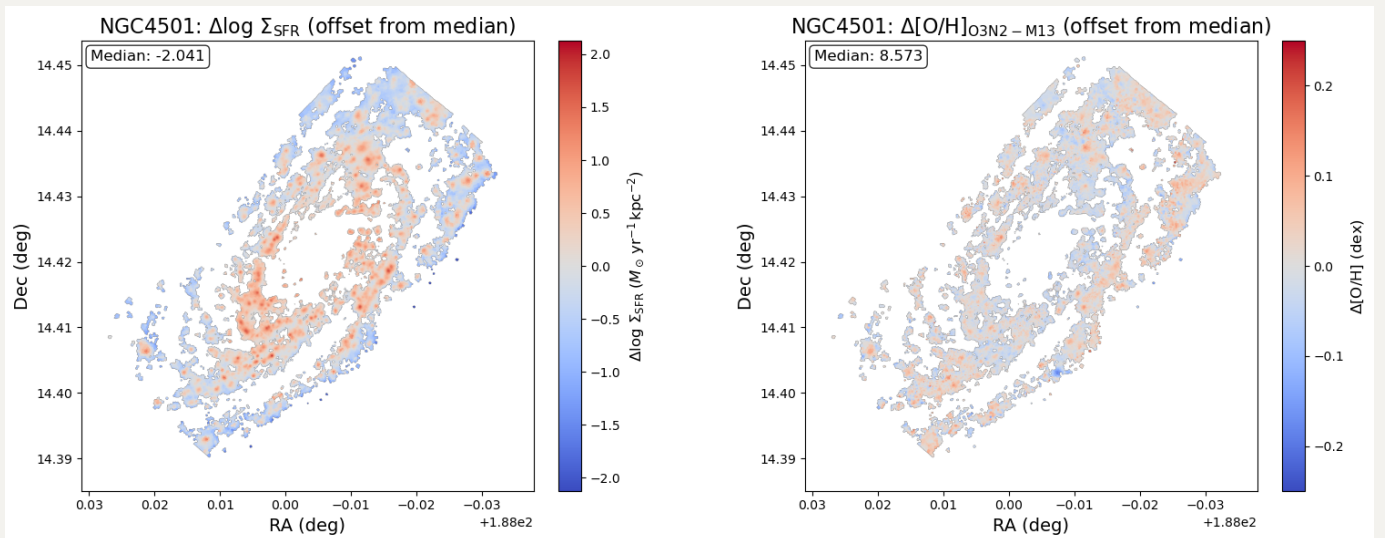


NGC4501

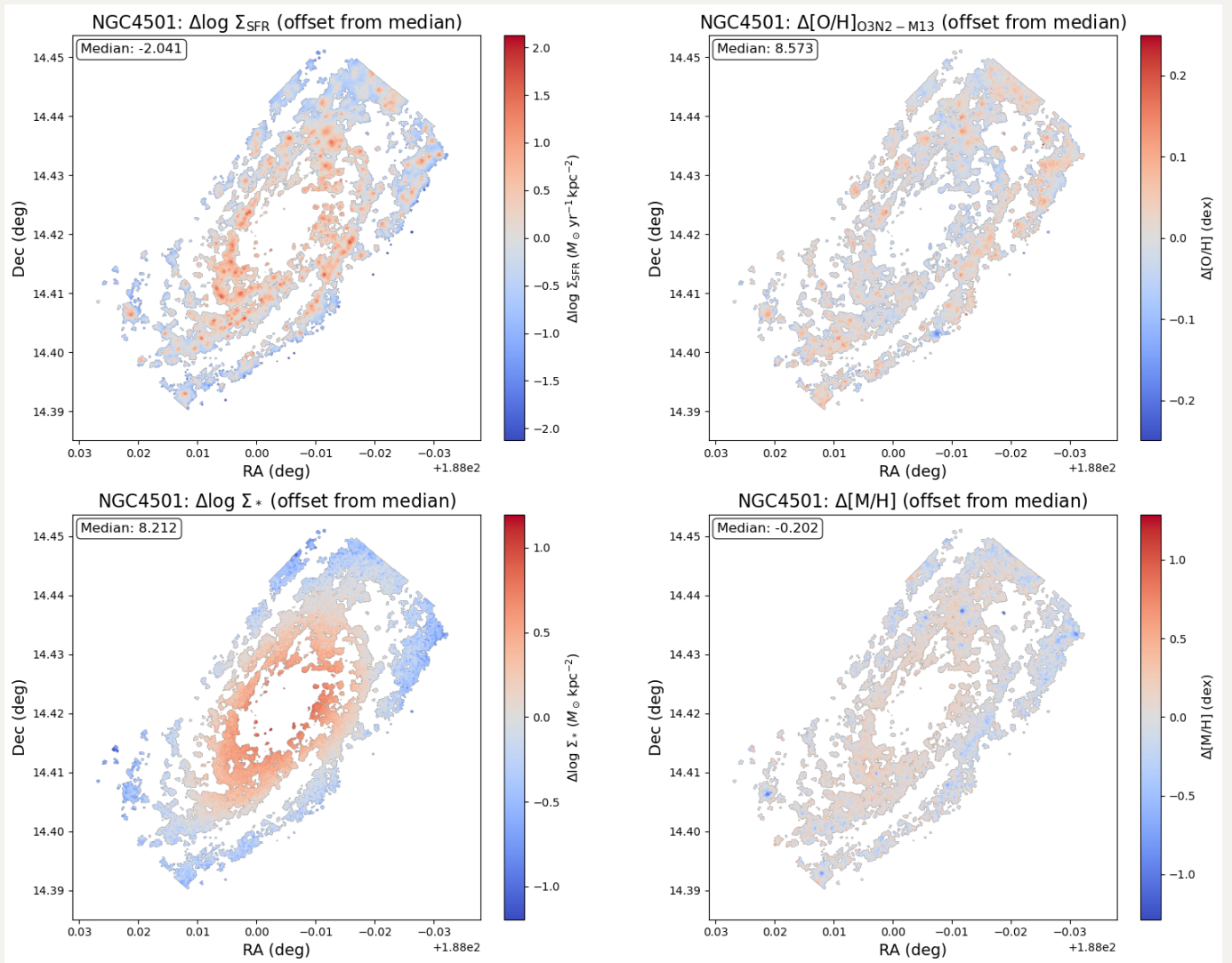
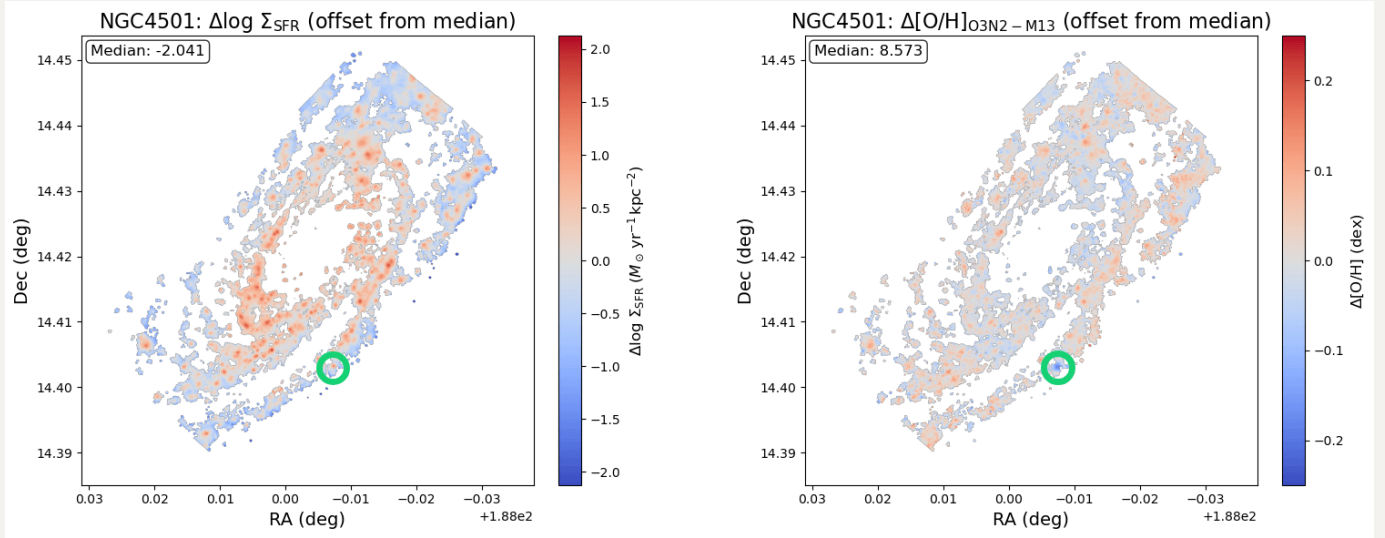
NGC4501 shows a clear positive correlation between Z_{gas} and Σ_{SFR} .



Here I try taking the median value as the standard (offset 0) and then perform the offset map to see how the Σ_{SFR} and Z_{gas} vary across the galaxy.

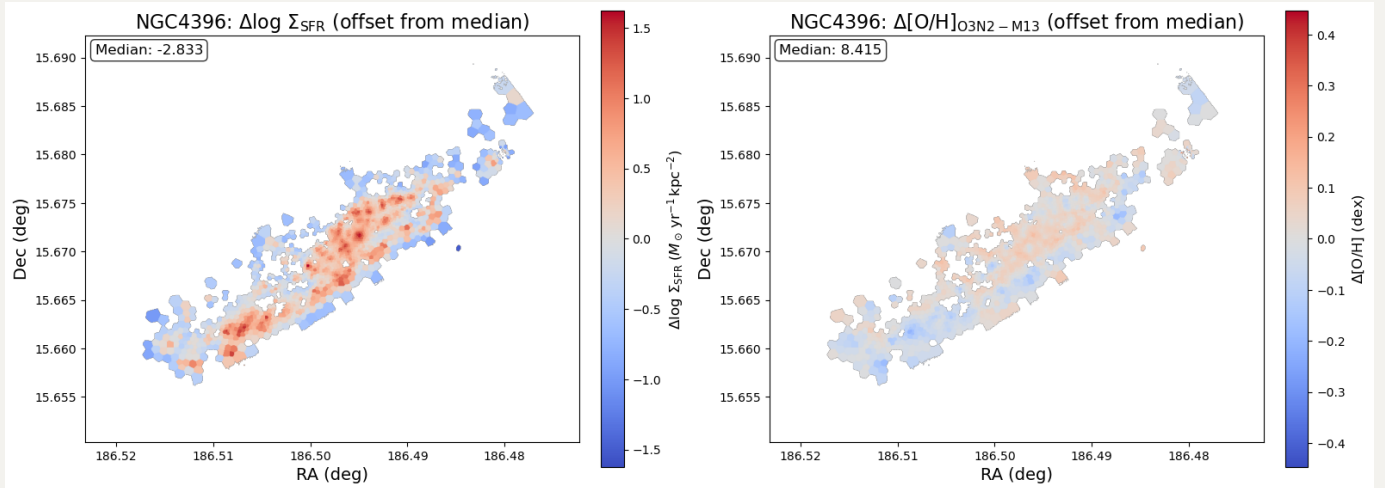
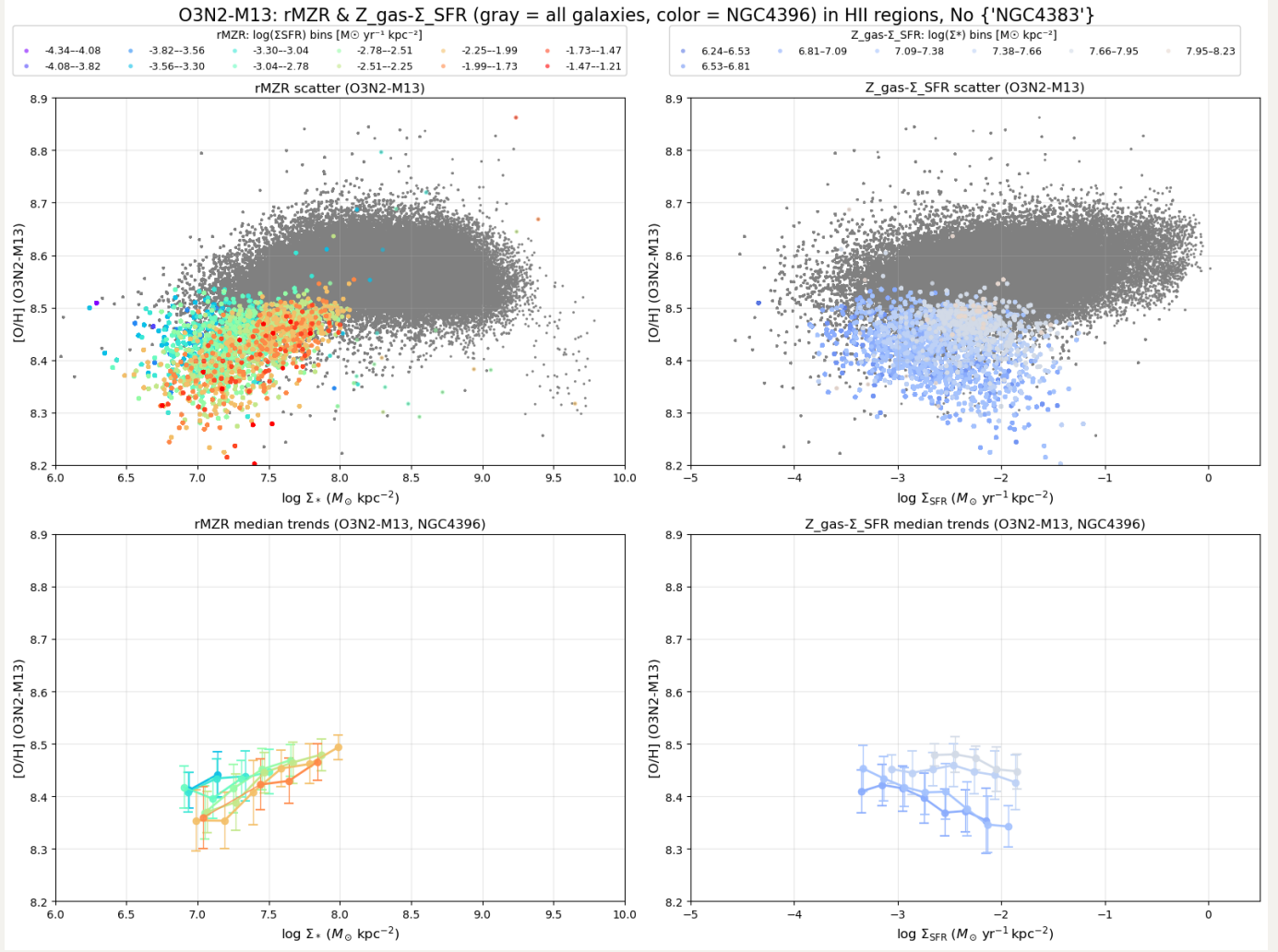


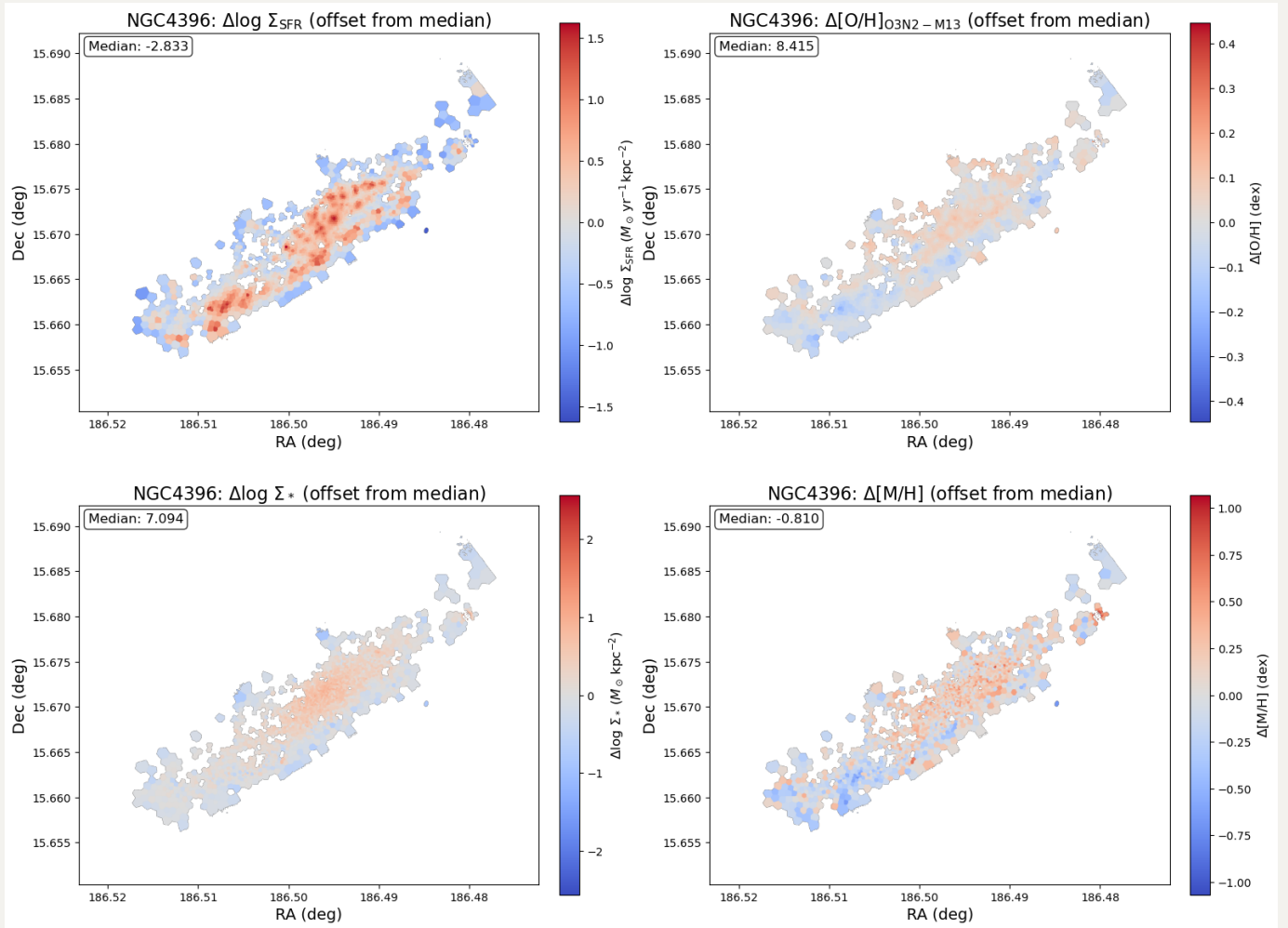
For NGC4501, we can see in general, redder on the LHS also shows redder on the RHS. However, we also notice that in the below green circle region, it shows spatial variation: the inverse correlation in this relatively low-mass HII regions.



NGC4396

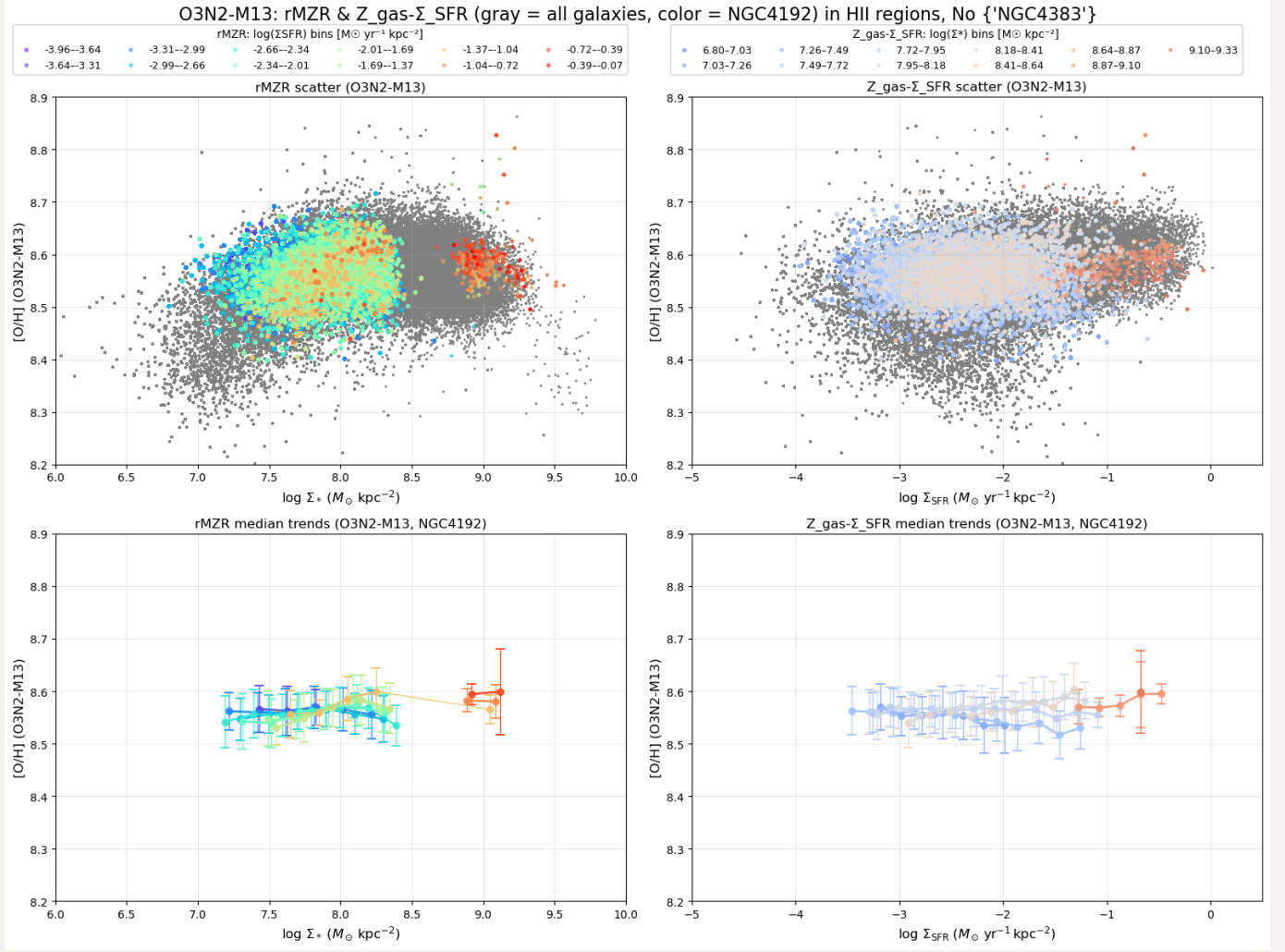
NGC4396 shows a clear negative correlation between Z_{gas} and Σ_{SFR} .



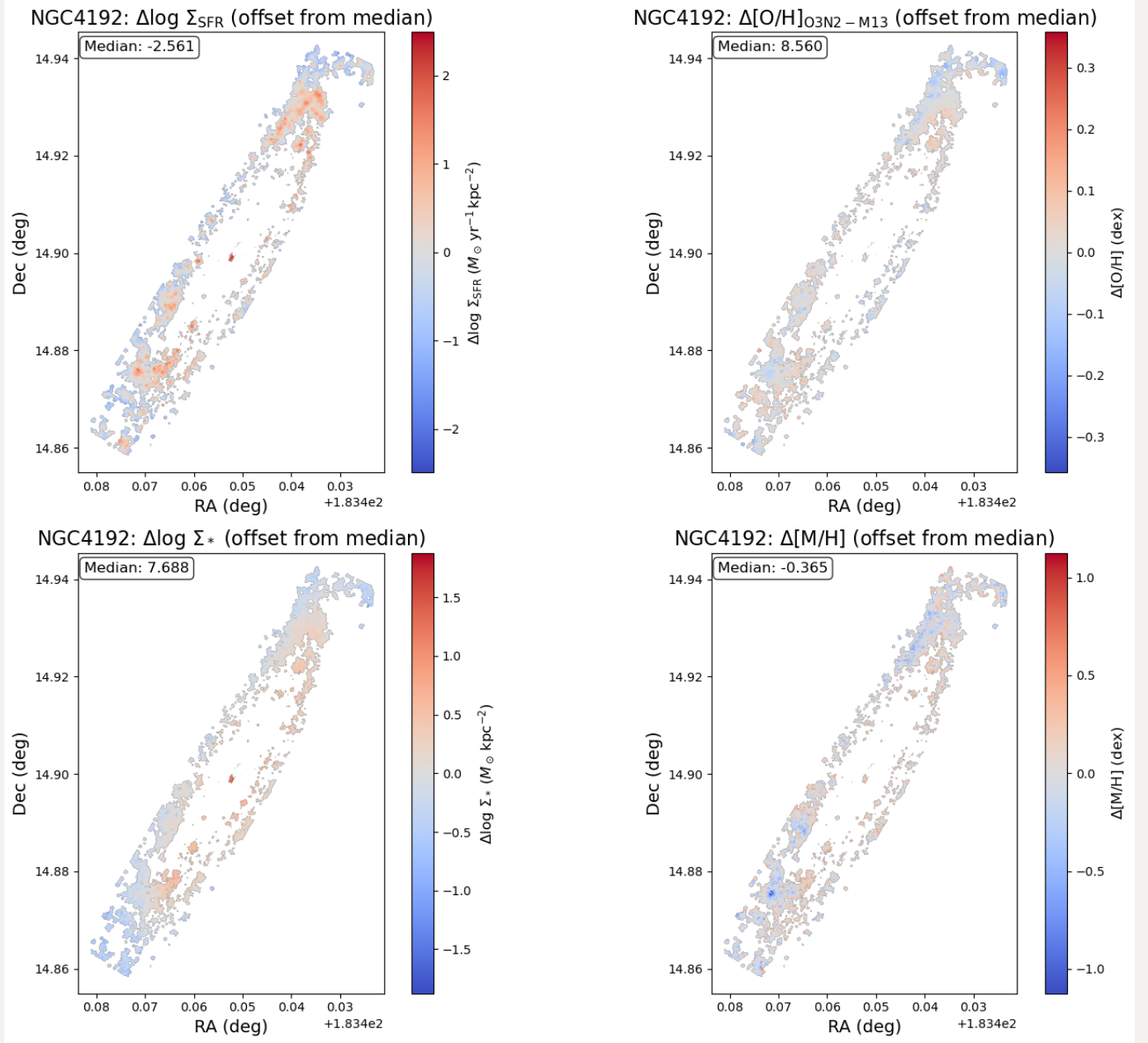


NGC4192

NGC4192 shows an ambiguous correlation between Z_{gas} and Σ_{SFR} ?



However, it shows a bit clearer inversion point at $\Sigma_{*} \sim 7.7$. So like the red regions ($\Sigma_{*} \gtrsim 7.7$) in $\Delta \log(\Sigma_{*})$ map show a positive correlation (redder in $\Delta \log(\Sigma_{\text{SFR}})$ also shows redder in Z_{gas}); while the blue regions ($\Sigma_{*} \lesssim 7.7$) in $\Delta \log(\Sigma_{*})$ map show a positive correlation (redder in $\Delta \log(\Sigma_{\text{SFR}})$ but shows bluer in Z_{gas}).



NGC4330

NGC4330 shows a negative correlation between Z_{gas} and Σ_{SFR} , though not that clear.

O3N2-M13: rMZR & Z_{gas}-Σ_{SFR} (gray = all galaxies, color = NGC4330) in HII regions, No {'NGC4383'}

